

ALGORITHM
Autumn 2024, Quiz 9, May 15
11:45-12:05 am

姓名：
學號：
系所：

1. Consider a modification of the rod-cutting problem in which, in addition to a price p_i for each rod, each cut incurs a fixed cost of c . The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. Give a dynamic-programming algorithm to solve this modified problem.

- (a) Suppose that each cut incurs a fixed cost of 2. Determine an optimal solution to cut a rod of length 10 with the following table of prices p_i for length $i = 1, 2, \dots, 10$.

length i	1	2	3	4	5	6	7	8	9	10
price p_i	1	3	6	6	10	9	8	13	8	12

- (b) Give a dynamic-programming algorithm to solve this modified problem.



Quize9-(a)

- $p[i]$: legth i 直接賣的價錢
 - $r[i]$: legth i 最好的價錢
 - $s[i]$: 最左邊那一刀的位置
- 變數定義、初始化: 1分

legth	0	1	2	3	4	5	6	7	8	9	10
$p[i]$	0	1	3	6	6	10	9	8	13	8	12
$r[i]$	0	1	3	6	6	10	10	11	14	14	18
$s[i]$	0	1	2	3	4	5	3	2	3	3	5

初始化 $r[0] = 0$ $s[0] = 0$

子問題最佳解: 2分
問題最佳解: 2分

$r[1] = \max\{ p[1] \} = 1, s[1] = 1$

$r[2] = \max\{ p[1]+r[1]-2, p[2] \} = 3, s[2] = 2$

$r[3] = \max\{ p[1]+r[2]-2, p[2]+r[1]-2, p[3] \} = 6, s[3] = 3$

$r[4] = \max\{ p[1]+r[3]-2, p[2]+r[2]-2, p[3]+r[1]-2, p[4] \} = 6, s[4] = 4$

$r[5] = \max\{ p[1]+r[4]-2, p[2]+r[3]-2, p[3]+r[2]-2, p[4]+r[1]-2, p[5] \} = 10, s[5] = 5$

$r[6] = \max\{ p[1]+r[5]-2, p[2]+r[4]-2, p[3]+r[3]-2, p[4]+r[2]-2,$
 $p[5]+r[1]-2, p[6] \} = 10, s[6] = 3$

$r[7] = \max\{ p[1]+r[6]-2, p[2]+r[5]-2, p[3]+r[4]-2, p[4]+r[3]-2,$
 $p[5]+r[2]-2, p[6]+r[1]-2, p[7] \} = 11, s[7] = 2$

$r[8] = \max\{ p[1]+r[7]-2, p[2]+r[6]-2, p[3]+r[5]-2, p[4]+r[4]-2,$
 $p[5]+r[3]-2, p[6]+r[2]-2, p[7]+r[1]-2, p[8] \} = 14, s[8] = 3$

$r[9] = \max\{ p[1]+r[8]-2, p[2]+r[7]-2, p[3]+r[6]-2, p[4]+r[5]-2,$
 $p[5]+r[4]-2, p[6]+r[3]-2, p[7]+r[2]-2, p[8]+r[1]-2, p[9] \} = 14, s[9] = 3$

$r[10] = \max\{ p[1]+r[9]-2, p[2]+r[8]-2, p[3]+r[7]-2, p[4]+r[6]-2, p[5]+r[5]-2,$
 $p[6]+r[4]-2, p[7]+r[3]-2, p[8]+r[2]-2, p[9]+r[1]-2, p[10] \} = 18, s[10] = 5$

[解] 說明最佳解切割方式: 1分

當長度為10時, 查表 $s[10]=5$, 表示會在5等分切一刀, 此時右邊長度剩下5,再查表 $s[5]=5$, 表示不用再切了, 所以在第5等分的位置切一刀, 會得到 $10 + 10 - 2 = 18$ 的最佳解。

Quiz9-(b) 4分

- p: 直接賣的價錢
- n: rod 長度
- c: 每刀成本

```
1  cut_rod(p, n, c)
2      int r[n] // 最好的價錢
3      int s[n] // 最左邊那一刀
4      r[0] = 0
5      for j = 1 to n {
6          int q = INT_MIN
7          for i = 1 to j
8              int cur_price = p[i] + r[j - i]
9              if (j - i != 0) // 扣除每刀成本
10                 cur_price -= c
11                 if(q < cur_price) // 如果價錢更高
12                     q = cur_price // 更新價錢
13                     s[j] = i // 記住最左邊那一刀
14             r[j] = q; // 記住最好的價錢
15     return r and s;
```

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(b) Give a dynamic-programming algorithm to solve this modified problem.

原 rod cutting 轉移式:

$$r(n) = \max_{1 \leq i \leq n} (p_i + r(n-i))$$

決定在邊切哪裡

考慮切的成本:

$$r(n) = \max \left(p_n, \max_{1 \leq i \leq n} (p_i + r(n-i) - c) \right)$$

整根 切一刀、扣成本、在邊長1
不切

(a) base case

i	0	1	2	3	4	5	6	7	8	9	10
$P(i)$	0	1	3	6	10	9	8	13	8	12	
$r(i)$	0	1	3	6	6	10	10	11	14	14	
$s(i)$	0	1	2	3	4	5	3	2	3	3	

$$r(1) = \max(P(1)) = 1$$

$$r(2) = \max\left(\frac{P(2)}{3}, \frac{P(1)+r(1)-2}{1+1-2=0}\right) = 3$$

$$r(3) = \max\left(\frac{P(3)}{6}, \frac{P(1)+r(2)-2}{1+3-2=2}, \frac{P(2)+r(1)-2}{3+1-2=2}\right) = 6$$

$$r(4) = \max\left(\frac{P(4)}{6}, \frac{P(1)+r(3)-2}{1+6-2=5}, \frac{P(2)+r(2)-2}{3+3-2=4}, \frac{P(3)+r(1)-2}{6+1-2=5}\right) = 6$$

$$r(5) = \max\left(\frac{P(5)}{10}, \frac{P(1)+r(4)-2}{1+6-2=5}, \frac{P(2)+r(3)-2}{3+6-2=7}, \frac{P(3)+r(2)-2}{6+3-2=7}, \frac{P(4)+r(1)-2}{6+1-2=5}\right) = 10$$

$$r(6) = \max\left(\frac{P(6)}{9}, \frac{P(1)+r(5)-2}{1+10-2=9}, \frac{P(2)+r(4)-2}{3+6-2=7}, \frac{P(3)+r(3)-2}{6+6-2=10}, \frac{P(4)+r(2)-2}{6+3-2=7}, \frac{P(5)+r(1)-2}{10+1-2=9}\right) = 10$$

$$r(7) = \max\left(\frac{P(7)}{8}, \frac{P(1)+r(6)-2}{1+10-2=9}, \frac{P(2)+r(5)-2}{3+10-2=11}, \frac{P(3)+r(4)-2}{6+6-2=10}, \frac{P(4)+r(3)-2}{6+6-2=10}, \frac{P(5)+r(2)-2}{10+3-2=11}, \frac{P(6)+r(1)-2}{9+1-2=8}\right) = 11$$

$$r(8) = \max\left(\frac{P(8)}{13}, \frac{P(1)+r(7)-2}{1+11-2=10}, \frac{P(2)+r(6)-2}{3+10-2=11}, \frac{P(3)+r(5)-2}{6+10-2=14}, \frac{P(4)+r(4)-2}{6+6-2=10}, \frac{P(5)+r(3)-2}{10+6-2=14}, \frac{P(6)+r(2)-2}{9+3-2=10}, \frac{P(7)+r(1)-2}{8+1-2=6}\right) = 11$$

$$r(9) = \max\left(\frac{P(9)}{8}, \frac{P(1)+r(8)-2}{1+14-2=13}, \frac{P(2)+r(7)-2}{3+11-2=12}, \frac{P(3)+r(6)-2}{6+10-2=14}, \frac{P(4)+r(5)-2}{6+5-2=9}, \frac{P(5)+r(4)-2}{10+6-2=14}, \frac{P(6)+r(3)-2}{9+3-2=10}, \frac{P(7)+r(2)-2}{8+3-2=9}, \frac{P(8)+r(1)-2}{13+1-2=12}\right) = 14$$

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(b) Give a dynamic-programming algorithm to solve this modified problem.

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$P(i)$	0	1	3	6	6	10	9	8	13	8	12
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$s(i)$	0	1	2	3	4	5	3	2	3	3	5

$$r(10) = \max \left(\frac{P(10)}{12}, \frac{P(1) + r(9) - 2}{1 + 14 - 2 = 13}, \frac{P(2) + r(8) - 2}{3 + 14 - 2 = 15}, \frac{P(3) + r(7) - 2}{6 + 11 - 2 = 15}, \frac{P(4) + r(6) - 2}{6 + 10 - 2 = 14}, \frac{P(5) + r(5) - 2}{10 + 10 - 2 = 18}, \frac{P(6) + r(4) - 2}{9 + 6 - 2 = 13}, \frac{P(7) + r(3) - 2}{8 + 6 - 2 = 12}, \frac{P(8) + r(2) - 2}{13 + 3 - 2 = 14}, \frac{P(9) + r(1) - 2}{8 + 1 - 2 = 7} \right)$$

當棍子長為10時，查表 $s[10]=5$ ，表示會在5等分處切一刀，此時右半部分長剩下5，繼續查表 $s[5]=5$ ，代表不用在切了。

→ 所以整個分割會在5等分處切一刀，可得最佳解價格為 $\frac{10}{\text{左半部分}} + \frac{10}{\text{右半部分}} - \frac{2}{\text{切割成本}} = 18$

原 rod cutting 轉移式：

$$r(n) = \max_{1 \leq i \leq n} (p_i + r(n-i))$$

決定左邊切哪裡

考慮切的成本：

$$r(n) = \max \left(\frac{P_n}{\text{整根不切}}, \max_{1 \leq i \leq n} (p_i + r(n-i) - c) \right)$$

切一刀、扣成本、左邊長 i

1. Consider a modification of the rod-cutting problem in which, in addition to a price p_i for each rod, each cut incurs a fixed cost of c . The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. Give a dynamic-programming algorithm to solve this modified problem.

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$$r(n) = \max \left(p_n, \max_{1 \leq i \leq n} (p_i + r(n-i) - c) \right)$$

整根 切一刀、扣成本、左邊長 i
不切

(b)

CutRod(p, n, c)

create 2 array r, s

$r[0] = 0$

for $j \leftarrow 1 \sim n$:

now $\leftarrow p[j]$

$s[j] = j$

for $i \leftarrow 1 \sim j$:

if now $< p[i] + r[j-i] - c$:

now $= p[i] + r[j-i] - c$

$s[j] = i$

$r[j] = \text{now}$

return r, s

1. Matrix-chain multiplication

- Show that a full parenthesization of an n -element expression has exactly $n - 1$ pairs of parentheses.
- Find an optimal parenthesization of matrix-chain product whose sequence of dimensions is $(10, 3, 12, 5, 50)$.

(a) 利用數學歸納法

basic:

2分 當 $n=1$ 時, 不需要任何一對括號即可使其產生 A_i 的乘積, 成立

induction step:

2分 當 $N=k$ 時, 先在乘積外圍加一層括號: $(A_1 A_2 \dots A_{k-1} A_k)$, 且存在一個數 m , 使得 $B = A_1 A_2 \dots A_m$, $C = A_{m+1} \dots A_k$, 改寫 $A_1 A_2 \dots A_k = B \times C$, 根據 strong induction hypothesis, B 需要 $m-1$ 對括號來表示乘積, 且 C 需要 $k-(m+1)$ 對括號來表示乘積, 將乘積數相加: $(m-1) + (k-(m+1)) + 1 = k-1$, 所以總共需要 $k-1$ 對括號來表示 $A_1 A_2 \dots A_k$, 故得證。
最外圍的括號

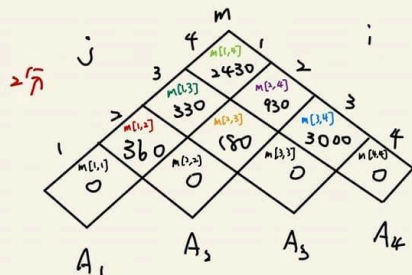
(b)

matrix	A_1	A_2	A_3	A_4
dimension	10×3	3×12	12×5	5×50

$$p_0 = 10 \quad p_1 = 3 \quad p_2 = 12 \quad p_3 = 5 \quad p_4 = 50$$

$m[i, j]$ 為 $A_i \dots A_j$ 的最少乘積數

$s[i, j]$ 為 $A_i \dots A_j$ 切在哪個位置



$$m[1, 2] = 10 \times 3 \times 12 = 360$$

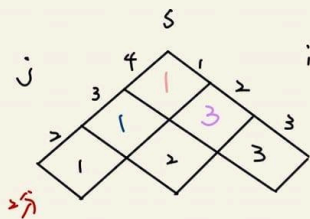
$$m[2, 3] = 3 \times 12 \times 5 = 180$$

$$m[3, 4] = 12 \times 5 \times 50 = 3000$$

$$m[1, 3] = \min \begin{cases} m[1, 1] + m[2, 3] + p_0 p_1 p_3 = 0 + 180 + 10 \times 3 \times 5 = 150 \\ m[1, 2] + m[3, 3] + p_0 p_2 p_3 = 360 + 0 + 10 \times 12 \times 5 = 960 \end{cases} = 150$$

$$m[2, 4] = \min \begin{cases} m[2, 2] + m[3, 4] + p_1 p_2 p_4 = 0 + 3000 + 3 \times 12 \times 50 = 4800 \\ m[2, 3] + m[4, 4] + p_1 p_3 p_4 = 180 + 0 + 3 \times 5 \times 50 = 930 \end{cases} = 930$$

$$m[1, 4] = \min \begin{cases} m[1, 1] + m[2, 4] + p_0 p_1 p_4 = 0 + 930 + 10 \times 3 \times 50 = 2430 \\ m[1, 2] + m[3, 4] + p_0 p_2 p_4 = 360 + 3000 + 10 \times 12 \times 50 = 9360 \\ m[1, 3] + m[4, 4] + p_0 p_3 p_4 = 930 + 0 + 10 \times 5 \times 50 = 2830 \end{cases} = 2430$$



查表 $s[1, 4]$ 切一刀在 A_1

查表 $s[2, 4]$ 切一刀在 A_3

查表 $s[2, 3]$ 切一刀在 A_2

$i = j = 3$, 停止查表

⇒ 最佳解為: $(A_1, ((A_2 \times A_3) \times A_4))$

1. Matrix-chain multiplication

- (a) Show that a full parenthesization of an n -element expression has exactly $n - 1$ pairs of parentheses.
- (b) Find an optimal parenthesization of matrix-chain product whose sequence of dimensions is $(10, 3, 12, 5, 50)$.

(a) * Chapter 15 P. 19

<一個有 n 個元素的算式，完全加上括號後恰好有 $n-1$ 對括號>
(n 個矩陣) (元素兩兩一組)

試證 $\forall n \geq 1$ ，任意 n 個矩陣的算式，在完全加上括號 (full parenthesization) 後恰有 $n-1$ 對括號

→ 利用強歸納法 (Strong Induction) 來證明

Base Case: $n=1$

當只存在一個矩陣時，不需要任何括號即可產生乘積，括號數 = 0，成立。

Inductive Hypothesis:

假設對於所有 $1 \leq j < n$ ， j 個元素完全加上括號後算式中存在 $j-1$ 對括號。

Inductive Step:

Let $n \geq 2$ ，根據定義， n 個元素的式子完全加上括號後似： $(B \cdot C)$

其中 B 為 k 元素的算式， C 為 $n-k$ 元素的算式， $1 \leq k \leq n-1$

由 I.H. 得知：

- B 有 $k-1$ 對括號
- C 有 $(n-k)-1$ 對括號

再加上最外層 $(B \cdot C)$ 有一對括號

可知總括號數為 $(k-1) + (n-k-1) + 1 = n-1$ 對括號

由歸納法得證。

1. Matrix-chain multiplication

(a) Show that a full parenthesization of an n -element expression has exactly $n - 1$ pairs of parentheses.

(b) Find an optimal parenthesization of matrix-chain product whose sequence of dimensions is $(10, 3, 12, 5, 50)$.

轉移式:

$$m(i, j) = \begin{cases} 0 & i=j \\ \min_{i \leq k < j} (m(i, k) + m(k+1, j) + P_{i-1}P_kP_j) & i < j \end{cases}$$

左邊最小乘法數 右邊最小乘法數 相乘乘法數

(b)

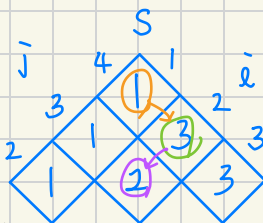
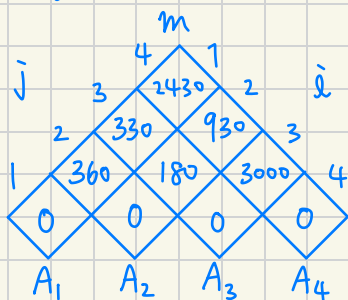
matrix	A_1	A_2	A_3	A_4
dim	10×3	3×12	12×5	5×50

<key> P 即維度序列 idx

P_0	P_1	P_2	P_3	P_4
10	3	12	5	50

$m[i, j]$ 為矩陣 $A_i \sim A_j$ 的最少乘積數

$S[i, j]$ 為矩陣 $A_i \sim A_j$ 的最佳分割位置



→ 根據結果 S 進行查表

$S(1, 4)$ 切在 $A_1 \rightarrow$ 剩 $A_2 \sim A_4$

$S(2, 4)$ 切在 $A_3 \rightarrow$ 剩 $A_2 \sim A_3$

$S(2, 3)$ 切在 $A_2 \rightarrow$ 完成查表

$\rightarrow (A_1 \times ((A_2 \times A_3) \times A_4))_{**}$

$$m(1, 3) = \min \left(\frac{m(1, 1) + m(2, 3) + P_0 P_1 P_3}{0 + 180 + 10 \times 3 \times 5} = 330, \frac{m(1, 2) + m(3, 3) + P_0 P_2 P_3}{360 + 0 + 10 \times 12 \times 5} = 960 \right), S(1, 3) = 1$$

$$m(2, 4) = \min \left(\frac{m(2, 2) + m(3, 4) + P_1 P_2 P_4}{0 + 3000 + 3 \times 12 \times 50} = 930, \frac{m(2, 3) + m(4, 4) + P_2 P_3 P_4}{180 + 0 + 3 \times 5 \times 50} = 930 \right), S(2, 4) = 3$$

$$m(1, 4) = \min \left(\frac{m(1, 1) + m(2, 4) + P_0 P_1 P_4}{0 + 930 + 10 \times 3 \times 50} = 2430, \frac{m(1, 2) + m(3, 4) + P_0 P_2 P_4}{360 + 3000 + 10 \times 12 \times 50} = 9360 \right)$$

$$\frac{m(1, 3) + m(4, 4) + P_0 P_3 P_4}{330 + 0 + 10 \times 5 \times 50} = 2830, S(1, 4) = 1$$